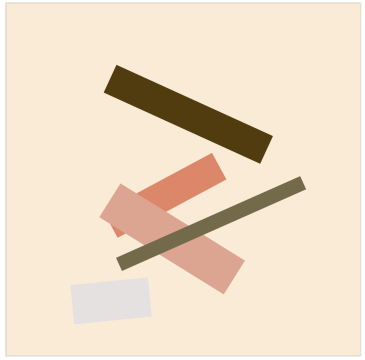
Make your first generative artwork

In less than 15 lines of code

by Raphaël de Courville

P5.JS ACTIVITY GUIDE



Adapted, with thanks, from this 2022 thread by Adam Genlight (@generativelight)

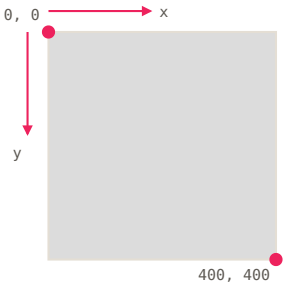
0 Open the editor

In a web browser go to **https://editor.p5js.org**

Delete the **`draw()`** function, leaving just this:

```
function setup() {
  createCanvas(400, 400);
}
```

Press the ▶ button. You'll see a grey square:



The coordinate system starts at the top-left corner.

1 Prepare the canvas

Add this line below **`createCanvas()`**:

```
background("antiquewhite");
```

Your code should look like this:

```
function setup() {
  createCanvas(400, 400);
  background("antiquewhite");
}
```

Now press the ▶ button again.

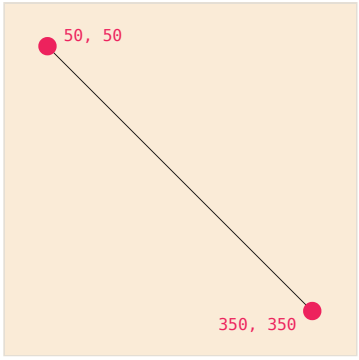
Your canvas is ready.

2 Draw one line

Next, add the following code just below the **`background()`** line:

```
line(50, 50, 350, 350);
```

`line()` is a function. It takes four arguments describing the start and end points of a line.



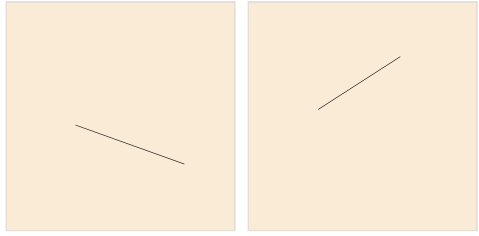
3 Leave it to chance

Swap all four numbers for **`random()`**:

```
line(random(50, 350), random(50, 350),
     random(50, 350), random(50, 350));
```

`random(50, 350)` returns a different number between 50 and 350 every time it is called.

Keep pressing ▶ and get a different line each time.



Same rules, different results.

4 Give it a body

Then add these lines immediately *above* the **`line()`**:

```
colorMode(HSL);
strokeCap(SQUARE);
strokeWeight(25);
stroke(random(0, 50), // hue
       random(0, 100), // saturation
       random(0, 100)); // lightness
```



YOUR TURN

Go to the p5.js reference and look up **`strokeCap()`** and **`strokeWeight()`**. Try different values for each.

5 Say it five times

Wrap the four lines from **`strokeCap()`** to **`line()`** in a **`while()`** loop:

```
let countDown = 5;
while (countDown > 0) {
  // your four lines go here
  countDown = countDown - 1;
}
```



YOUR TURN

Change the **`countDown`** number to make **10** lines, or **25**, or **200**! Use whole numbers only.

The whole sketch

```
function setup() {
  createCanvas(400, 400);
  background("antiquewhite");
  colorMode(HSL);
  let countDown = 5;
  while (countDown > 0) {
    strokeCap(SQUARE);
    strokeWeight(25);
    stroke(random(0, 50), // hue
           random(0, 100), // saturation
           random(0, 100)); // lightness
    line(random(50, 350), random(50, 350),
         random(50, 350), random(50, 350));
    countDown = countDown - 1;
  }
}
```

scan to run it

YOUR TURN

Can you spot the small difference? Try editing the code to make the output look more like the one on the cover.

Hint: it has to do with the stroke weight.

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Solution: use `strokeWeight(random(10, 50));`